

Mohammad Ghaffary

Senior Backend Engineer · Python · Distributed Systems

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SUMMARY

Senior Backend Engineer with 8 years of experience designing, scaling and modernizing Python systems, most recently at **Nobitex, Iran's largest cryptocurrency exchange (10M+ users)**. Hands-on with **Django/DRF, FastAPI, event-driven microservices with Kafka and RabbitMQ, PostgreSQL performance tuning, Docker and Kubernetes**. I turn slow, risky manual processes into automated, well-tested ones, and I mentor other engineers along the way.

SKILLS

Backend	Python, Django, DRF, FastAPI, Celery, SQLAlchemy, Pydantic, REST/OpenAPI, pytest (TDD), SQL, Bash
Distributed systems	Microservices, Apache Kafka, RabbitMQ, event-driven architecture, idempotency, concurrency control
Data	PostgreSQL (query tuning, indexing, locking), Redis, MongoDB, ClickHouse, Alembic
DevOps & tools	Docker, Kubernetes, CI/CD, Linux, Prometheus, Grafana, Git, n8n, LLM API integration

EXPERIENCE

Freelance Backend Engineer — QueueLess (Remote) · *event-driven virtual-queue SaaS* 2026 – Present

- Designed a microservices architecture (Identity, Queue, Business, Notification, Analytics) with a database per service and versioned Kafka domain events, published by idempotent producers keyed per queue for ordering.
- Built the core Queue Service with FastAPI, SQLAlchemy 2.x and PostgreSQL. Row-level locks and atomic compare-and-set state transitions guarantee **no duplicate tickets** and **exactly one winner** when concurrent actions race.

Senior Software Engineer — Nobitex (Tehran) · *largest crypto exchange in Iran, 10M+ users* Aug 2023 – May 2026

- Re-architected coin and blockchain-network integration to be configuration-driven, cutting the time to add a new coin from **~3 hours to ~5 minutes** with **no deployment**.
- Built a compatibility layer that mapped new Django REST Framework responses to legacy formats, enabling an incremental migration with **zero breaking changes** for clients.
- Designed a deposit-simulation harness that shadow-tested refactored money-handling code against legacy code and logged any differences, making the refactoring **~20% faster** and safer.
- Mentored a mid-level engineer to deliver a Telegram bot that lets product owners list new tokens themselves, removing engineers from routine token listings and **freeing the team for core work**.
- Tuned PostgreSQL queries and ORM access patterns on critical paths, reducing database hits by **up to 50%**.

Backend Engineer (Django) — Learnwise (Remote) · *Misalearn language-learning app* Jul 2022 – Aug 2023

- Cut client requests for dictionary updates by **95%** by replacing per-word polling with encrypted JSON bundles generated periodically with Celery Beat.
- Built a reusable Django module for metadata generation, data validation and key control that integrates with multiple modules and became a shared building block across the application.
- Reached **100% API documentation coverage** with auto-generated Swagger/OpenAPI docs, overriding its schema classes to support every custom field type and django-filter endpoint.
- Developed and published an n8n community node, and integrated LLM-generated content into the product.

Backend Engineer & Data Analyst — Farafan (Tehran) · *telecom & IT, 300+ employees* Jun 2018 – Jun 2022

- Built the Django backend of **VirusPaad**, a SaaS that scans uploaded files with **40+ antivirus engines** and serves both a web app and on-site kiosks.
- Created a centralized control system for **40+ VMs** running antivirus engines, using RPyC for remote command execution to distribute scans and collect results.
- Integrated Samba and FTPS for file transfer between the platform and scanning VMs, and aggregated per-engine verdicts to flag infected files.
- Radio Patrol**: automated **~80%** of manual telecom reporting with scheduled jobs that turn large-scale mobile network data in ClickHouse into KPIs in PostgreSQL, visualized in Grafana dashboards.

EDUCATION

M.Sc. in Information Technology Engineering — K. N. Toosi University of Technology, Tehran · **Languages**: English (professional working proficiency), Persian (native), Kurdish (native)